By (5) & (6) putting into $\Delta u = 0$, we have

$$\Delta u = (u_{xx}) + (u_{yy}) = (4\xi u_{\xi\xi} + 4\sqrt{\xi}\, u_{\xi\eta} + u_{\eta\eta} + 2u_{\xi}) + (u_{\eta\eta}) = 0 \Rightarrow$$

(7) $$\Delta u = 4\xi u_{\xi\xi} + 2u_{\eta\eta} + 4\sqrt{\xi}\, u_{\xi\eta} + 2u_{\xi} = 0$$ is NOT

a Laplace eq-n of $\nabla \cdot \nabla u(\xi, \eta) = 0$. Hence, the given transformation fails to preserve the harmonicity.

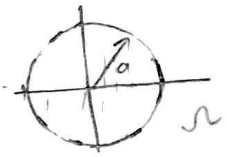
In a nutshell,

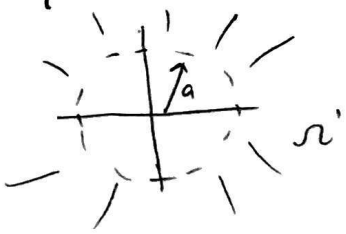
$$u(x,y) \text{ is harmonic in } \Omega \not\Rightarrow u(\xi,\eta) \text{ is harmonic in } \Omega'$$

where Ω' is obtained by some transformation. ∎

Prob. 4. Show that if $u(r,\theta)$ is a harm f-n in a circle, $r^2 < a^2$, then, $u(r', \theta')$ is harmonic in the <u>exterior</u> circle $r^2 > a^2$, where

(1) $$\xi r = a^2 \qquad \eta = \theta$$





Sol-n: Let $u(r,\theta)$ be harmonic in $\Omega = \{r^2 < a^2\}$ i.e.

(1) $u(r,\theta) \in C^2(\Omega)$ & $\Omega = \{(r,\theta) : r^2 < a^2\}$

(2) $\Delta u = 0$

Then the used transformation:

(3) $$\xi = \frac{a^2}{r} \qquad \eta = \theta$$

Let's follow the hint in the introduction of this HW 10:

(i) $|J| \neq 0$, verification:

(4) $$J = \begin{vmatrix} \eta_r & \eta_\theta \\ \xi_r & \xi_\theta \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ -\frac{a^2}{r^2} & 0 \end{vmatrix} = \frac{a^2}{r^2} \neq 0 \Rightarrow$$

the given trans-n is OK as long as $r \neq 0$ & $a \neq 0$.

(ii) $\Delta u = 0$ must stay unaffected by the trans-n:

(5) $$\Delta u = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} = 0 \Rightarrow$$

in polar

(6) $$\Delta u(\xi, \eta) = \frac{1}{\xi} \cdot \frac{\partial}{\partial \xi}\left[\xi \cdot \frac{\partial u}{\partial \xi}\right] + \frac{1}{\xi^2} \cdot \frac{\partial^2 u}{\partial \eta^2}$$ must be got.

Starting with (5)

$$\Delta u(r,\theta) = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot \frac{\partial}{\partial r}\left(u(\xi,\eta)\right)\right] + \frac{1}{r^2}\frac{\partial}{\partial \theta}\left[\frac{\partial}{\partial \theta}\left[u(\xi,\eta)\right]\right]$$

$$= \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot (u_\xi \cdot \xi_r + u_\eta \cdot \eta_r)\right] + \frac{1}{r^2} \cdot \frac{\partial}{\partial \theta}\left[u_\xi \xi_\theta + u_\eta \eta_\theta\right] =$$

$$= \frac{1}{r} \cdot \left\{1 \cdot (u_\xi \xi_r + u_\eta \eta_r) + r \cdot \frac{\partial}{\partial r}\left[u_\xi \xi_r + u_\eta \eta_r\right]\right\} + \frac{1}{r^2} \cdot \left\{u_{\xi\xi} \xi_\theta^2 + 2u_{\xi\eta} \xi_\theta \eta_\theta + u_{\eta\eta} \eta_\theta^2 + u_\xi \xi_{\theta\theta} + u_\eta \eta_{\theta\theta}\right\} =$$

$$= \frac{u_\xi \xi_r + u_\eta \eta_r}{r} + \left(u_{\xi\xi} \xi_r^2 + 2u_{\xi\eta} \xi_r \eta_r + u_{\eta\eta} \eta_r^2 + u_\xi \xi_{rr} + u_\eta \eta_{rr}\right) +$$

$$+ \frac{u_{\xi\xi} \xi_\theta^2 + 2u_{\xi\eta} \xi_\theta \eta_\theta + u_{\eta\eta} \eta_\theta^2 + u_\xi \xi_{\theta\theta} + u_\eta \eta_{\theta\theta}}{r^2} \quad (=)$$

[can replace r's later to keep simple]

$$= \frac{u_\xi \cdot \left(-\frac{a^2}{r^2}\right) + u_\eta \cdot (0)}{r} + \left[u_{\xi\xi}\left(-\frac{a^2}{r^2}\right)^2 + 2u_{\xi\eta}\left(\frac{-a^2}{r^2}\right)\cdot 0 + u_{\eta\eta}\cdot(0)^2 + u_\xi\left(\frac{2a^2}{r^3}\right) + u_\eta \cdot (0)\right]$$

$$+ \frac{u_{\xi\xi}(0)^2 + 2u_{\xi\eta}(0)\cdot(1) + u_{\eta\eta}(1)^2 + u_\xi(0) + u_\eta(0)}{r^2} =$$

$$= \left(-\frac{a^2}{r^3}\right)\cdot u_\xi + \frac{a^4}{r^4}u_{\xi\xi} + \left(\frac{2a^2}{r^3}\right)u_\xi +$$

$$+ \left(\frac{1}{r^2}\right)u_{\eta\eta} = \frac{a^4}{r^4}u_{\xi\xi} + \frac{1}{r^2}u_{\eta\eta} + \frac{a^2}{r^3}u_\xi \Rightarrow \text{ That is}$$

(7) $$\Delta u = \frac{a^4}{r^4}u_{\xi\xi} + \frac{a^2}{r^3}u_\xi + \frac{1}{r^2}\cdot u_{\eta\eta} = 0 \Big\} \cdot \frac{r^4}{a^4} \Rightarrow \boxed{u_{\xi\xi} + \frac{r}{a^2}u_\xi + \frac{r^2}{a^4}u_{\eta\eta} = 0} \quad (7')$$

Now is this (7) identical with (6)?

(8) $$\Delta u(\xi,\eta) = \frac{1}{\xi}\cdot\frac{\partial}{\partial\xi}\left[\xi\cdot\frac{\partial u}{\partial\xi}\right] + \frac{1}{\xi^2}\cdot\frac{\partial^2 u}{\partial\eta^2} =$$

$$= \frac{1}{\xi}\cdot\left[1\cdot u_\xi + \xi\cdot u_{\xi\xi}\right] + \frac{1}{\xi^2}\cdot u_{\eta\eta} \Rightarrow$$

(8') $$\boxed{\Delta u(\xi,\eta) = u_{\xi\xi} + \frac{1}{\xi}u_\xi + \frac{1}{\xi^2}\cdot u_{\eta\eta}} \Rightarrow \left[\xi = \frac{a^2}{r},\ \eta = \theta\right]$$

$$\Delta u = 0 \Rightarrow u_{\xi\xi} + \frac{r}{a^2}u_\xi + \frac{r^2}{a^4}u_{\eta\eta} = 0 \quad \Big\} \cdot \frac{a^4}{r^4}$$

Forcing to $\Rightarrow$ make similar to (7):

$$\frac{a^4}{r^4}u_{\xi\xi} + \frac{a^4}{r^{\cancel{4}}}\cdot\frac{\cancel{r}}{a^2}u_\xi + \frac{a^{\cancel{4}}}{r^4}\cdot\frac{\cancel{r^2}}{\cancel{a^4}}u_{\eta\eta} = 0 \Rightarrow$$

(9) $$\frac{a^4}{r^4}u_{\xi\xi} + \frac{a^2}{r^3}\cdot u_\xi + \frac{1}{r^2}u_{\eta\eta} = 0$$

Of course (7') $\equiv$ (8') & (7) $\equiv$ (9), thus Laplace eq-n does not change & given trans-n keeps $u(\xi,\eta)$ harmonic. ∎